



Balancing performance and sustainability in integrated CO₂ capture and methanation via NiCeO₂-anionic clay hybrid materials

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ABSTRACT

Integrated CO₂ capture and methanation (ICCM) provides a direct route for converting captured CO₂ into synthetic methane, but its large-scale implementation is limited by material stability and sustainability. Here, we report NiCeO₂/anionic clay hybrid materials, where the Mg-Al oxide derived from the layered double hydroxide (LDH) clay acts as an adsorbent matrix for NiCeO₂ catalytic nanoparticles. The hybrids exhibit enhanced CO₂ uptake, efficient low-temperature operation (300 °C) with 100% CH₄ selectivity, and excellent cyclic stability under ICCM conditions. Although the NiCeO₂/LDH hybrids show lower methane productivity per unit mass of total material than NiCeO₂ alone, they significantly improve catalyst utilization, producing up to seven times more CH₄ per unit mass of NiCeO₂. Operando XANES reveals a reversible Ni²⁺/Ni⁰ redox transition under alternating CO₂/H₂ atmospheres, demonstrating that the active Ni phase dynamically adjusts its oxidation state during CO₂ adsorption–hydrogenation cycles. Life cycle assessment demonstrates a markedly reduced environmental footprint for the hybrid materials compared with NiCeO₂ for the same methane output, highlighting NiCeO₂/LDH hybrids as sustainable and scalable platforms for integrated CO₂ capture and conversion.

1. Introduction

The reduction of anthropogenic CO₂ emissions represents one of the most pressing challenges for achieving sustainable energy systems and low-carbon chemical production. Beyond mitigation, the valorisation of captured CO₂ into useful fuels and chemicals offers a promising pathway to close the carbon cycle while enabling large-scale energy storage [1]. In this context, and given the technical and economic limitations associated with hydrogen storage and transport, CO₂ methanation has emerged as an attractive route for carbon utilization and energy storage [2]. Methane is a safe, clean-burning energy carrier that can be readily integrated into existing natural gas infrastructure, enabling rapid

deployment across power, heating, and transportation sectors [3].

Integrated CO₂ capture and methanation (ICCM) using hybrid materials that combine adsorption and catalytic functionalities provides a direct route for converting captured CO₂ into CH₄ [4–7]. A key factor governing the development of efficient ICCM technologies is the performance of these hybrid materials, which must simultaneously exhibit high CO₂ uptake and high catalytic activity. Dual-functional materials (DFMs) are therefore typically designed by integrating an adsorbent with a catalytic phase and may additionally incorporate a support and/or promoter to enhance stability and reactivity [8]. Common adsorbents are based on alkali and alkaline-earth metals such as Ca, Na, Mg, and K. The catalytic component is selected according to the targeted CO₂

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conversion reaction, and for methanation Ni and Ru are the most widely employed active metals. In many formulations, oxide supports such as Al_2O_3 are introduced to improve dispersion and mechanical robustness, while small amounts of noble metals are often added as promoters to enhance reducibility and catalytic performance [4–6]. Several recent comprehensive reviews summarize the current state of the art, performance, and remaining challenges of DFM, including Ni-based systems [8–10].

Despite their widespread use, conventional alkali and alkaline-earth-based adsorbents suffer from important limitations, including slow adsorption kinetics, volatilization or sintering at elevated temperatures, and, in some cases, limited long-term stability under cyclic operation [5]. These drawbacks can compromise the overall ICCM efficiency, highlighting the need to explore alternative adsorption materials that combine fast CO_2 uptake, thermal robustness, and long-term stability under reaction conditions. In this context, layered double hydroxides (LDHs), offer an attractive platform due to their compositional flexibility and tunable surface chemistry. These lamellar materials consist of mixed divalent and trivalent metal cations arranged in positively charged hydroxide layers, with charge-compensating anions and water molecules located in the interlayer spaces [11]. Their modular composition enables systematic control of basicity and surface properties, allowing CO_2 adsorption over a broad temperature range [12–14]. Consequently, LDH-based materials have been widely investigated in various CO_2 capture configurations, where their structural adaptability and process compatibility highlight their potential for advanced capture technologies [15].

Incorporation of catalytically active metals for CO_2 hydrogenation, such as Ni or Ru, has further expanded the functionality of LDH-derived materials beyond simple CO_2 adsorbents. These systems have been investigated under conventional catalytic methanation conditions [16,17], chemical looping schemes [18], as well as in cyclic ICCM processes. Most ICCM studies to date have focused on Ni-containing LDHs in which nickel is structurally incorporated into the brucite-like layers during synthesis [19–24]. After thermal activation, these materials transform into dual-function systems capable of simultaneously capturing CO_2 and catalyzing its hydrogenation to methane. In contrast, LDH-based hybrid architectures in which the catalytic phase is deposited onto preformed LDH supports have been explored far less [25–27]. The development of such hybrid systems, where adsorption and catalytic functions can be independently tuned while preserving close spatial coupling, represents a promising strategy for next-generation capture–conversion materials.

In this work, we present a hybrid ICCM material that integrates an LDH adsorbent with NiCeO_2 catalytic nanoparticles for CO_2 methanation. The system operates efficiently at relatively low temperatures and exhibits excellent cyclic stability, while avoiding the use of costly noble metals. Importantly, life cycle assessment demonstrates that although the hybrid material delivers lower ICCM performance than NiCeO_2 alone, it offers a markedly reduced environmental footprint, making it the more attractive option for large-scale implementation. These results highlight the potential of NiCeO_2 /LDH hybrids as sustainable and scalable platforms for integrated CO_2 capture and conversion.

2. Materials and methods

2.1. Materials synthesis

2.1.1. Synthesis of NiCeO_2 nanocatalyst

The NiCeO_2 catalyst was prepared using a previously reported Schiff-base complex route [28–30]. The choice of this synthesis method and the particular NiCeO_2 catalyst was guided by its favorable characteristics for catalytic applications, including high specific surface area, small particle size, high synthesis yield, and good CO_2 methanation performance demonstrated in our previous work [30]. The synthesis proceeds through three consecutive steps: (i) preparation of the monometallic Ce

(III)–L and Ni(II)–L complexes; (ii) dissolution of 5.1 g of Ce(III)–L and 1.5 g of Ni–L in 40 mL of oleylamine, followed by pyrolysis at 180 °C under Ar for 2 h, during which nanoparticles (NPs) are formed in solution; and (iii) extraction of the NPs by successive addition of 2.5 mL of methanol per 5 mL of solution. The remaining supernatant was subjected to two additional NP extraction cycles (2.5 mL each), and the NiCeO_2 catalyst used in this study corresponds to the third extraction cycle [30]. The resulting nanoparticles were finally calcined at 450 °C for 1 h with 5 °C/min ramp rate to form a powder. The calcined catalyst exhibited a specific surface area of 120 $\text{m}^2 \text{g}^{-1}$, measured by N_2 physisorption, and a Ni content of 12 wt%, determined by Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES).

2.1.2. Synthesis of LDH

LDH with a theoretical structure $\text{Mg}_{0.75}\text{Al}_{0.25}(\text{OH})_2(\text{CO}_3)_{0.125} \cdot n \text{H}_2\text{O}$ was prepared by the co-precipitation method aiming 3 g of product. 1 M magnesium nitrate and aluminum nitrate solutions were prepared using magnesium nitrate hexahydrate (Sigma Aldrich, 99%) and aluminum nitrate nonahydrate (Carlo Erba, 98–102%) salts and deionized water (18.2 $\text{M}\Omega\cdot\text{cm}$). The solutions were mixed in 3:1 = Mg:Al ratio (solution A). 2 M sodium hydroxide solution was prepared from solid NaOH pellets (Sigma Aldrich, 97%) and deionized water (solution B). The quantity of OH^- ions was calculated considering LDH stoichiometry and the final solution pH corresponding to 10. Solution A was poured into solution B under magnetic stirring. Anion exchange was performed adding 0.2 M Na_2CO_3 (Sigma Aldrich, 99.5%) solution to the suspension (x2 in excess with respect to LDH stoichiometry). The suspension was aged with continuous magnetic stirring for 24 h under ambient conditions. The precipitate was washed utilizing deionized water several times to eliminate the excess of Na^+ and NO_3^- ions by centrifugation (8000 rpm for 5 min, e.g. 9946 $\times g$). The obtained powders were then dried overnight at 60 °C and crushed in agate mortar. The specific surface area of mixed oxide derived from LDH (calcined at 400 °C) was 280 $\text{m}^2 \text{g}^{-1}$. Upon thermal treatment under reducing conditions, the LDH structure collapses and transforms into a Mg–Al mixed oxide, commonly referred to in the literature as a layered double oxide (LDO) [31]. Despite this structural transformation, in the present work we retain the abbreviation “LDH” of the precursor to refer to the material even after thermal treatment.

2.1.3. Synthesis of the hybrid NiCeO_2 /LDH

Hybrid NiCeO_2 /LDH catalysts, denoted as NCL_x (where x represents the overall dilution factor of the NiCeO_2 phase, defined as the total mass ratio of $\text{NiCeO}_2 + \text{LDH}$ to NiCeO_2), were prepared by ball milling a NiCeO_2 catalyst with the LDH support for 1 h 40 min at 400 rpm using a Retsch Mixer Mill MM 400 equipped with a 5 mL zirconia jar and two 7 mm zirconia balls. The parent NiCeO_2 catalyst (containing 12 wt% Ni) was first diluted in LDH using a 1:3 NiCeO_2 :LDH mass ratio. Specifically, 80 mg of NiCeO_2 were mixed with 240 mg of LDH and ball-milled at 400 rpm for 1 h 40 min, yielding a homogeneous powder referred to as NCL_4 . Further dilutions were achieved by successive ball-milling steps with LDH. For the synthesis of NCL_6 , 120 mg of NCL_4 were mixed with 120 mg of LDH and ball-milled under identical conditions (400 rpm, 1 h 40 min). Similarly, NCL_{16} was prepared by ball milling 120 mg of NCL_6 with 120 mg of LDH at 400 rpm for 1 h 40 min. All samples were collected as homogeneous powders and used without further treatment. This step-wise dilution strategy enables a controlled decrease in NiCeO_2 content while preserving the physicochemical properties of the active phase. The absence of detectable zirconia contamination arising from abrasion of the zirconia milling media was verified by XPS (Fig. S1a) and SEM-EDX analyses.

2.2. Materials characterization

2.2.1. Structural, textural, and redox characterization

The materials were analyzed by X-ray diffraction (XRD) using a

Brüker D8 Advance X-ray diffractometer (Cu K α irradiation). Nitrogen physisorption volumetry was employed to measure the specific surface areas of the materials using an ASAP 2020 Micromeritics device. The specific surface area was calculated using the Brunauer-Emmett-Teller (BET) theory. Scanning electron microscopy (SEM) combined with energy-dispersive X-ray spectroscopy (EDX) was performed using a Zeiss GeminiSEM 500 microscope. Annular dark field scanning transmission electron microscopy (STEM) was performed with an image-corrected Titan ETEM G2 (FEI) operated at 300 kV and equipped with an energy-dispersive X-ray (EDX) spectrometer. For sample preparation, the powder was crushed and dispersed in ethanol through ultrasonication, dropped onto a holey carbon-coated 200-mesh Cu grid, and dried using a lamp. X-ray photoelectron spectroscopy (XPS) was performed using a PHI Genesis instrument (base pressure in the 2×10^{-9} mbar) using a monochromatic Al X-ray source. The analysis spot diameter was 100 μ m. Electron and ion neutralizer guns were turned on during spectra acquisition to compensate surface charging. Temperature-programmed experiments were carried out with 50 mg of material into a homemade setup coupled with a mass spectrometer to monitor the different masses: $m/z = 2$ for H $_2$, $m/z = 15$ and 16 for CH $_4$, $m/z = 28$ for CO, and $m/z = 44$ for CO $_2$. Prior to any CO $_2$ -involving experiment, a reducing pretreatment at up to 500 °C (5 °C/min, plateau 30 min) under pure hydrogen flow (40 mL/min) was performed. Subsequently, hydrogen was replaced with helium (30 mL/min) for 30 min. Afterwards, the reactor was cooled down to 50 °C. For temperature-programmed desorption of CO $_2$ (CO $_2$ -TPD), the CO $_2$ pre-adsorption step consisted in flowing 1% CO $_2$ /He at 30 mL/min for 30 min at 50 °C, followed by helium flowing (30 mL/min) for 30 min to remove residual CO $_2$ from the stream along with CO $_2$ weakly adsorbed on the material surface. A temperature ramp of 10 °C/min was then conducted up to 500 °C. CO $_2$ -TPD cycles consisted in repeating the same procedure on the same sample several times without any intermediate step. For temperature-programmed hydrogenation of pre-adsorbed CO $_2$ (CH $_4$ -TPSR), 1% CO $_2$ /He was flowed (30 mL/min) through the reactor for 30 min at 50 °C. Then, the gas was switched to 10% H $_2$ /He for 30 min at a flow rate of 30 mL/min. Subsequently, a temperature ramp of 10 °C/min was applied until reaching 500 °C.

2.2.2. Cyclic CO $_2$ hydrogenation experiments

For these experiments, 150 mg of catalyst were used. Prior to the cycles, a pretreatment at 400 °C (5 °C/min) under pure H $_2$ with a flow rate of 40 mL/min for 1 h was carried out. Then, the reactor was cooled down to 300 °C. For each cycle, a 7.5% $_{v/v}$ CO $_2$ /N $_2$ flow was introduced into the reactor at a rate of 120 mL/min for 3 min, followed by a N $_2$ purging step at 120 mL/min for 1.5 min. Subsequently, the sample was exposed to a 33% $_{v/v}$ H $_2$ /N $_2$ flow at 120 mL/min for 3 min, followed by another N $_2$ purging step for 1 min. This process was repeated successively over at least 10 cycles. Methane and CO productions were calculated by integrating the corresponding peaks for each cycle, then normalizing by the total mass of materials used in the reactor or by the mass of the NiCeO $_2$ catalyst alone. The use of model CO $_2$ /N $_2$ mixtures enables one to assess the intrinsic performance of the materials under well-controlled conditions. However, additional flue gas components, such as H $_2$ O and O $_2$, are known to influence both CO $_2$ capture and catalytic behaviour [32] and should therefore be considered in future studies aimed at evaluating the suitability of these materials for practical flue gas applications.

2.2.3. Operando X-ray absorption spectra under ICCM conditions

Ni K-edge X-Ray Absorption Spectra (XAS) were collected at the ID24-DCM beamline [33] of the European Synchrotron Radiation Facility (ESRF) during beamtime CH-7064 [33,34]. Monochromatic beam was obtained with Si(111) Double Crystal Monochromator (DCM) while a double Si mirror was set at 2.8 mrad for harmonic rejections. Three ionization chambers of 15 cm length were employed for measuring incoming beam (I_0), beam transmitted by the sample (I_1) and beam

transmitted by the reference metal foil (I_2). I_0 was filled with 1 bar of N $_2$ while I_1 and I_2 with 70 mbar of Ar and 930 mbar of N $_2$. The sample was loaded as a loose powder in the reactor and it was kept in place using a carbon window [35]. Heat was provided by a resistive element while gases were dosed through calibrated Bronkhorst Mass Flow Controllers. Ni K-edge spectra were collected in fluorescence mode using a Si diode detector with a 40 μ m Al filter to reduce the noise originating from Ce fluorescence lines. XAS spectra were collected in continuous mode in the 8.1–9.4 keV energy range with 0.3 eV/point energy resolution and 5 ms/point integration time. The XAS spectra were energy aligned to the spectrum of I_0 previously calibrated to Ni metal foil. Spectra were then edge jump normalized and background subtracted using Python based Larch script [36]. FT-EXAFS spectra were obtained by Fourier Transformation of k^2 -weighted $\chi(k)$ experimental spectra in the 2.5–14 $\text{\AA}^{-1}$ k -range. Linear Combination Fit was performed with Larch-based python scripts using NiO and Ni spectra as references. Analysis was performed in the energy region $-100/+100$ eV respect to the edge position. References fit quantities were constrained to be positive and their values to sum to unity.

2.3. Life cycle assessment methodology

The LCA was performed using SimaPro v.9.6.0.1 PhD, in combination with the ReCiPe 2016 Midpoint (H) v1.08/World (2010) H impact assessment method using the International Organization for Standardization (ISO) 14,040 and 14,044 standards [37]. The LCA methodology follows the synthesis and mixing procedures described in Section 2.1.2, using the exact experimental quantities and operating conditions. Given the lack of representative datasets for the materials used in the synthesis, custom life cycle inventories (LCIs) were developed for all key materials based on publicly available data and experimental parameters. Accordingly, separate LCAs were performed for NiCeO $_2$ and LDH, while for the NCL hybrids, the energy demand associated with the ball-milling mixing step was included alongside the impacts of NiCeO $_2$ and LDH, weighted according to their respective mass fractions. NCL $_4$ was initially modeled by combining NiCeO $_2$ and LDH at a 1:3 mass ratio, including the electricity consumption of the ball-milling step (0.25 kWh). Subsequently, NCL $_8$ was generated by mixing the previously obtained NCL $_4$ with additional LDH at a 1:1 mass ratio, again accounting for the corresponding mixing energy. The same sequential approach was applied to obtain NCL $_{16}$, with each mixing step incorporating the associated electricity demand. A detailed list of the materials, including their precise quantities involved in the synthesis of NiCeO $_2$ and LDH, is provided in Table S1. The functional unit (FU) was defined as 1 kg of material produced. The system boundaries were set to “Cradle-to-Gate,” encompassing all stages from the extraction of raw materials through synthesis and energy use up to the point where the catalyst is ready for application. Downstream stages such as distribution and use were excluded. The present study focuses exclusively on the environmental impacts associated with DFM production. The use phase was excluded due to the lack of reliable long-term durability data. Therefore, the results should be interpreted as a comparative assessment of DFM synthesis rather than a complete evaluation of the overall ICCU process. The cut-off method was applied, and geographically consistent (France-based) datasets were used where available. A Monte Carlo uncertainty analysis (1000 iterations, ISO 14044) was performed to assess the robustness of the comparative results. Further details on inventory construction, modeling assumptions, and uncertainty propagation are provided in the Supporting Information.

3. Results and discussion

3.1. Structural characterization and CO₂ adsorption–hydrogenation properties

3.1.1. Structural morphological and surface characterization

3.1.1.1. Structural characterization by XRD. The crystallographic structures of the hybrid materials and their individual components, after reduction at 400 °C for 1 h, are deduced from the XRD diffractograms presented in Fig. 1. The XRD pattern of the reduced LDH corresponds to a cubic MgO structure (Fm3[−]m, periclase, PDF 45–496), consistent with literature reports on annealed Mg–Al hydrotalcites with similar molar ratios [38]. The broad peak indicates that the oxide is poorly crystallized, and no distinct Al₂O₃-related reflections are observed in the diffractograms. The lattice parameter derived from the XRD data is slightly smaller than that of pure MgO (4.18 Å from the (200) reflection vs 4.21 Å for cubic MgO), suggesting partial incorporation of Al³⁺ into the MgO lattice [39], while Scherrer analysis of the Mg(Al)O (220) reflection at 63° yields an estimated crystallite size of 4.2 nm (see Table S2), consistent with the broad diffraction peaks. The broad peak around 35 degrees observable for LDH and NCL₁₆ corresponds to a defective spinel structure [40]. The XRD patterns of the NiCeO₂ catalysts predominantly show the cubic CeO₂ phase, with smaller diffraction peaks corresponding to NiO, in agreement with our previous reports [30]. The diffractograms of the NCL hybrid materials are a weighted contribution from the XRD patterns of the LDH and NiCeO₂ constituents, with NiCeO₂-related diffraction peaks decreasing as the composition moves from NCL₄ to NCL₁₆. No significant changes are observed in the diffraction peak positions or widths, suggesting that the hybrid material is a physical mixture of the two constituents rather than a new mixed phase. However, in the NCL₈ sample, new narrow diffraction lines corresponding to the MgAl₂O₄ spinel structure (PDF 00–008-0234) are observed. The CeO₂ crystallite sizes reported in Table S2, determined from the CeO₂ (111) reflection using the Scherrer equation, range from 5.3 to 5.5 nm for both the parent NiCeO₂ catalyst and all NCL hybrid materials, indicating that the mechanochemical ball-milling procedure

does not alter the size of the CeO₂ nanocrystals.

On the basis of the XRD results obtained for the full NCL series (NCL₄–NCL₁₆), NCL₈ was selected as a representative intermediate composition for further surface and morphological investigations. This sample provides a balanced contribution of both NiCeO₂ and LDH phases and is therefore particularly relevant for probing the structural and interfacial features governing the CO₂ adsorption/hydrogenation behaviour. Moreover, as will be shown later, NCL₈ exhibits an optimal balance between environmental burden and ICCM performance, further justifying its selection for detailed characterization. Accordingly, detailed surface chemical analysis was performed on NCL₈ by XPS.

3.1.1.2. Surface characterization by XPS. Survey spectra (Fig. S1a) confirm the presence of elements from both constituents of the hybrid material, namely Ni and Ce from the catalyst and Mg and Al from the LDH phase. For NCL₈, the surface composition determined by XPS (Table S3) corresponds to 11.6 at.% NiCeO₂ and 88.4 at.% LDH. Notably, the surface Mg/Al ratio calculated from Mg 2p and Al 2p decreases from 3.0 in pristine LDH (75% Mg, 25% Al) to 2.0 in NCL₈, indicating significant Al enrichment at the NCL₈ surface. The Ni 2p, Ce 3d, Mg 1 s, and Al 2p spectra reveal oxidized surface species, as expected for ex situ measurements where the sample has been exposed in air. Because Mg core-level chemical shifts are small and the samples are insulating, Mg oxidation states cannot be reliably distinguished from binding energies alone. Instead, the modified Auger parameter (α'), defined as the sum of the Mg KLL kinetic energy and Mg 2p binding energy, is unaffected of charging effects [41,42]. The α' values of 1230.6 eV for NCL₈ and 1230.1 eV for pristine LDH indicate a modified Mg chemical environment in NCL₈, consistent with the observed Mg/Al surface redistribution observed in the quantitative analysis. However, definitive assignment of Mg chemical state remains difficult due to large variation in reported reference values: early studies [41,43] assign $\alpha' \approx 1230.8$ eV to MgO, whereas more recent work reports $\alpha' = 1231.3$ eV for MgO and 1230.4 eV for Mg(OH)₂ [42], rendering MgO attribution less conclusive in the present case.

3.1.1.3. Morphological characterization by SEM and STEM.

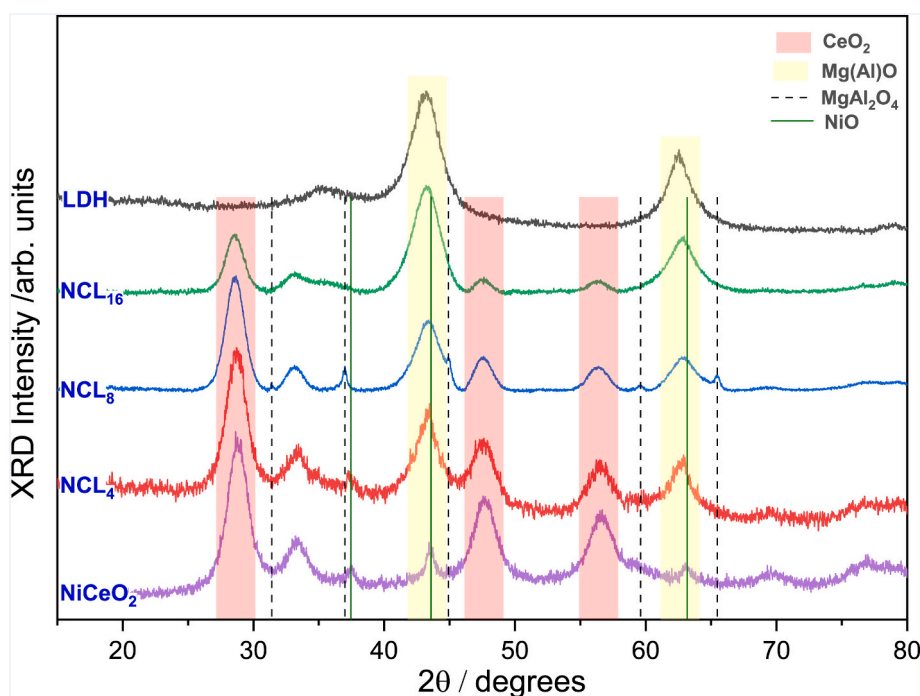


Fig. 1. X-ray diffraction (XRD) patterns of NiCeO₂, pristine LDH, and NCL hybrid materials (NCL₄, NCL₈, and NCL₁₆) after reduction at 400 °C for 1 h. Shaded regions highlight reflections assigned to cubic CeO₂ (red) and MgO periclase (yellow), while green and black lines indicate NiO and MgAl₂O₄ phases, respectively.

characterization of NCL₈ (Fig. 2a) reveals an aggregated morphology composed of irregular, micron-sized LDH particles forming the darker background and decorated with finer nanoscale particles exhibiting varying degrees of agglomeration. At low magnification, the material appears as loosely packed agglomerates, whereas higher-magnification secondary-electron (SE) images show a rough and porous texture arising from interconnected nanocrystallites. Combined secondary and energy-selective backscattered electron imaging (SE + ESB) displays pronounced compositional contrast, indicative of a heterogeneous spatial distribution of the NiCeO₂ and LDH phases. Local EDX analyses acquired from selected regions corroborate this heterogeneity, with NiCeO₂ contributions ranging from minor surface decoration (~6%) to locally enriched areas reaching ~56%, while LDH remains the predominant phase across most regions. The overall EDX composition confirms an LDH-rich material, in agreement with XRD and XPS results. These observations point to incomplete intermixing of the two components, resulting in the coexistence of NiCeO₂-rich and LDH-rich domains at the micrometer scale.

High-angle annular dark-field (HAADF) STEM combined with EDX elemental mapping provides further insight into the nanoscale organization of NCL₈ (Fig. 2b and Fig. S2). The HAADF image reveals nanoparticles with sizes ranging from 7 to 35 nm. Closer inspection shows that the larger particles consist of agglomerates of smaller crystallites anchored on the LDH support, consistent with the small NiCeO₂ particle size achieved by the selected synthesis method [28,30]. EDX mapping shows strong spatial correlation between Ni and Ce, confirming intimate intermixing within the same nanoscale domains rather than segregated Ni and Ce phases, thereby validating the strategy of using pre-formed catalytic nanoparticles instead of post-impregnation of individual precursors. In contrast, Mg and Al exhibit a broader and more homogeneous distribution, consistent with the continuous LDH phase.

3.1.2. Temperature-programmed CO₂ desorption and hydrogenation characterization

3.1.2.1. CO₂-TPD analysis. We next compare the CO₂ adsorption properties of LDH, NiCeO₂, and the hybrid NCL₈ materials based on their CO₂-TPD profiles. As shown in Fig. S3, all samples exhibit a main desorption maximum in the low-temperature region (≈ 120 – 140 °C), characteristic of weak basic adsorption sites [44]. The peak maximum appears at 127 °C for LDH and shifts to higher temperature for NCL₈ (137 °C), indicating a moderate strengthening of CO₂ binding upon formation of the hybrid material, likely originating from interfacial interactions between the NiCeO₂ and LDH phases. Beyond the main desorption feature, all profiles display broad high-temperature tails extending up to ~ 450 – 500 °C, reflecting a wide distribution of adsorption strengths and the presence of more strongly bound carbon-

containing species. This pronounced tailing points to significant surface heterogeneity across all materials. Quantitatively, the total CO₂ uptake follows the order NCL₈ ($0.173 \text{ mmol CO}_2 \text{ g}^{-1}$) > NiCeO₂ ($0.131 \text{ mmol CO}_2 \text{ g}^{-1}$) > LDH ($0.116 \text{ mmol CO}_2 \text{ g}^{-1}$), demonstrating an enhanced CO₂ adsorption capacity for the hybrid material compared with both parent phases.

3.1.2.2. CH₄-TPSR analysis. The CO₂ hydrogenation behaviour of NiCeO₂ and NCL₈ was further investigated by temperature-programmed surface reaction (CH₄-TPSR) following CO₂ pre-adsorption (Fig. S4). As will be shown later, the LDH showed minor hydrogenation activity and is not included. Upon switching from CO₂ to H₂ and increasing the temperature, CO₂ desorbs while CH₄ forms in parallel. NCL₈ shows a delayed onset of CH₄ formation compared to NiCeO₂, with CH₄ evolution starting around 250 °C and peaking near 300 °C. The delayed onset of CH₄ formation for NCL₈ indicates reduced methanation activity at lower temperatures. In contrast, NiCeO₂ produces CH₄ at lower temperatures, reflecting its higher intrinsic hydrogenation activity. Simultaneously, the CO₂ profiles reveal broader and more intense desorption features for NCL₈ compared to NiCeO₂, extending over a wider temperature range. This behaviour suggests that a significant fraction of the CO₂-derived species adsorbed on NCL₈ is less readily hydrogenated and therefore desorbs before being converted to CH₄.

Quantitative CH₄-TPSR analysis (Fig. 3a) reveals marked differences between NiCeO₂ and NCL₈. The NCL₈ sample produces significantly less CH₄ and releases more CO₂ than NiCeO₂, corresponding to a lower CO₂ conversion efficiency (50% for NCL₈ vs 90% for NiCeO₂), estimated from the ratio of desorbed CH₄ to the total desorbed carbon species (CH₄ + CO₂). Comparison of the total gas-phase carbon recovery obtained from CO₂-TPD and CH₄-TPSR experiments shown in Fig. 3b, further highlights distinct behaviour between the two materials. For NCL₈, the total amount of carbon recovered is very similar in both experiments. In contrast, NiCeO₂ releases twice as much carbon during CH₄-TPSR (sum of CO₂ and CH₄) compared to CO₂-TPD. This suggests that, on NiCeO₂, a significant fraction of CO₂ is not fully recovered during inert desorption but remains stored at the surface, as also observed earlier for Re/CeO₂ [27]. In the presence of H₂, these stored species are hydrogenated due to their proximity with the reaction sites, and leave the surface as CH₄. As we have shown before [30,45,46], these species are most likely stable surface carbonates, which also serve as reactive intermediates in CO₂ methanation. In the case of NCL₈, CO₂ adsorption appears to be largely reversible in the TPD experiment, resulting in a similar carbon balance between CO₂-TPD and CH₄-TPSR, and indicating a lower contribution of strongly retained carbonate species.

CO₂-TPD shows that NCL₈ retains substantial amounts of adsorbed CO₂, while CH₄-TPSR indicates that CH₄ formation reaches a maximum around 300 °C. Below 300 °C, a significant fraction of CO₂ desorbs

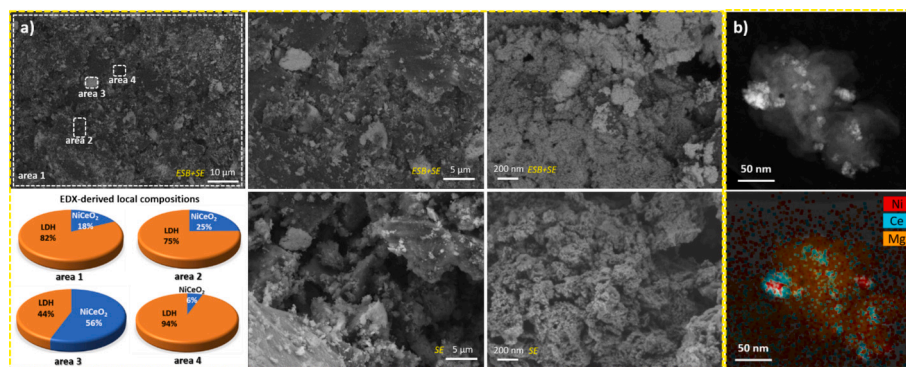


Fig. 2. Microscopy characterization of the NCL₈ sample. (a) SEM images acquired at different magnification scales in secondary electron (SE, bottom) and combined secondary electron and energy-selective backscattered (SE + ESB, top) modes. Corresponding EDX analyses from selected regions are shown as pie charts, illustrating the local chemical composition. SE imaging emphasizes the three-dimensional morphology, whereas SE + ESB imaging enhances the compositional contrast between different phases. (b) HAADF-STEM image and corresponding EDX elemental maps.

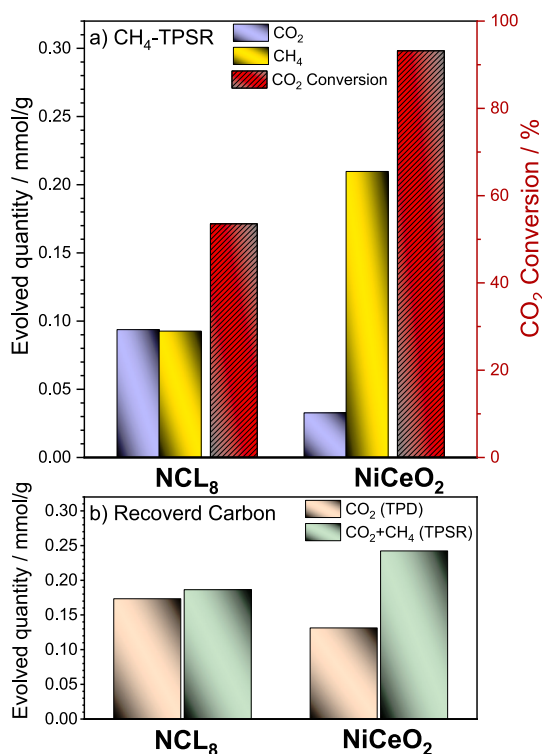


Fig. 3. (a) Quantitative CH₄-TPSR results after CO₂ pre-adsorption on NCL₈ and NiCeO₂, showing CO₂ desorption, CH₄ formation, and % CO₂ conversion to CH₄. (b) Gas-phase carbon recovered from CO₂-TPD and CH₄-TPSR experiments.

without conversion, whereas higher temperatures favour methanation kinetics. Accordingly, 300 °C represents a good compromise between CO₂ retention and hydrogenation activity of the NCL material, and was therefore selected as the operating temperature for the ICCM tests presented in the next section.

3.2. Integrated CO₂ capture and conversion performance of NCL materials

The ICCM performance of the NCL materials was evaluated through more than ten consecutive isothermal CO₂ adsorption–hydrogenation cycles at 300 °C. Fig. 4 presents the CO₂ and CH₄ outlet signals during a

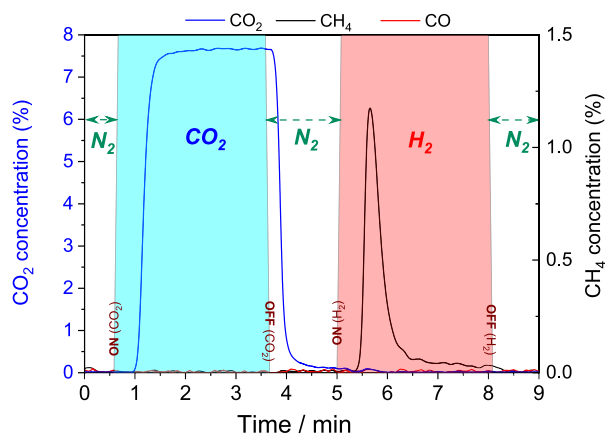


Fig. 4. Representative isothermal ICCM cycle over NCL at 300 °C showing outlet CO₂ and CH₄ concentrations during sequential CO₂ adsorption (7.5% CO₂/N₂, 120 mL min^{−1}, 3 min), N₂ purge (120 mL min^{−1}, 1.5 min), H₂ hydrogenation (33% H₂, 120 mL min^{−1}, 3 min), and final N₂ purge (120 mL min^{−1}, 1 min).

representative ICCM cycle. During the CO₂ adsorption step (cyan background), the CO₂ concentration rapidly increases and reaches saturation within approximately 90 s, demonstrating fast CO₂ capture by the material. Upon switching to H₂, CH₄ is detected at the reactor outlet, and the hydrogenation reaction is essentially completed within ~60 s, indicating rapid conversion of the stored CO₂. No detectable CO is observed during the hydrogenation step, excluding a significant contribution from the reverse water–gas shift reaction, and no unreacted CO₂ is released. These results confirm the efficient and selective integrated CO₂ capture and conversion to CH₄ over NCL at 300 °C.

The CO₂ and CH₄ evolution profiles recorded during 11 consecutive ICCM cycles at 300 °C for NCL₄, NCL₈, and NCL₁₆ hybrids are shown in Fig. S5. For comparison, the results for NiCeO₂ and pure LDH are given in Fig. S6. For all three NCL materials and across all cycles, the CO₂ and CH₄ concentrations during the CO₂ adsorption and H₂ hydrogenation steps, respectively, follow trends very similar to those observed for the cycle presented in Fig. 4. The nearly identical signal profiles over the 11 cycles demonstrate high cyclic stability and reproducibility for all NCL materials, with particularly consistent CH₄ production profiles. This behaviour extends over longer cycling, as shown for the NCL₈ sample in Fig. S7. The peak-normalized CH₄ signal intensity exhibits a moderate decrease with cycling over 50 CO₂/H₂ cycles, however, the CH₄ yield estimated from the integrated CH₄ profile area remains essentially constant, indicating preserved overall methane productivity.

Comparison of the CH₄ signal across the different samples in Figs. S5 and S6, shows differences in CH₄ production, indicating a strong composition dependence of the ICCM performance. Fig. 5 further illustrates this by comparing the average CH₄ production over 11 consecutive CO₂ adsorption–hydrogenation cycles for the three NCL hybrid materials, alongside their individual LDH and NiCeO₂ components. The CH₄ production is reported both per total mass of material loaded in the reactor (mmol CH₄ g^{−1} cycle^{−1}) and per mass of NiCeO₂ present in each composite (mmol CH₄ g^{−1} NiCeO₂ cycle^{−1}). When normalized to the total material mass (Fig. 5, left y-axis), the NCL hybrids display intermediate CH₄ yields (0.129–0.201 mmol CH₄ g^{−1} cycle^{−1}), situated between the negligible activity of pristine LDH (0.019 mmol CH₄ g^{−1} cycle^{−1}) and the substantially higher yield of NiCeO₂ (0.286 mmol CH₄ g^{−1} cycle^{−1}). The superior gravimetric CH₄ productivity of NiCeO₂ highlights its primary role as the active methanation phase [46], as well as the favorable CO₂ adsorption properties of ceria [47]. In the hybrid materials, partial replacement of NiCeO₂ by the LDH sorbent necessarily reduces the fraction of catalytically active sites per unit mass of composite.

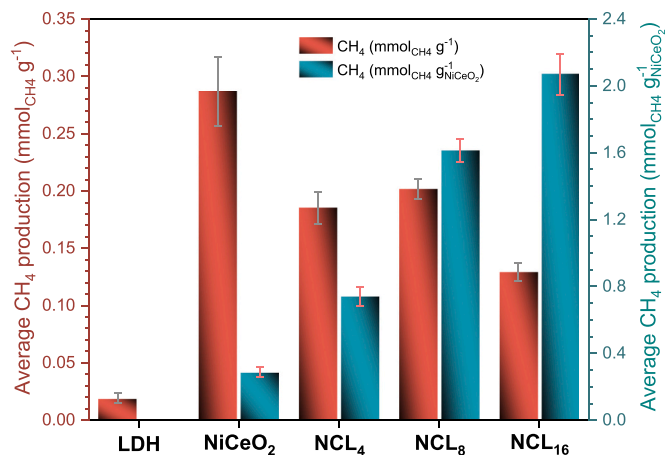


Fig. 5. CH₄ production per cycle averaged over 11 consecutive CO₂ adsorption/hydrogenation cycles for LDH, NiCeO₂, and the NCL hybrid materials (NCL₄, NCL₈, and NCL₁₆). CH₄ yields are expressed both per total mass of material loaded in the reactor (mmol CH₄ g^{−1}, red bars) and per mass of NiCeO₂ catalyst (mmol CH₄ g^{−1} NiCeO₂, blue bars). Error bars represent the standard deviation over the 11 cycles.

Consequently, the apparent CH₄ production expressed per gram of material decreases, primarily as a result of this compositional dilution rather than an intrinsic loss of catalytic efficiency.

When this dilution effect is removed by normalizing CH₄ production to the NiCeO₂ content (Fig. 5, right y-axis), a markedly different trend becomes evident. Under this normalization, all NCL hybrids outperform the standalone NiCeO₂ catalyst, with NCL₁₆ displaying a CH₄ productivity per gram of NiCeO₂ nearly seven times higher than that of NiCeO₂ alone. This demonstrates that, although NiCeO₂ is the active phase responsible for CO₂ hydrogenation, its utilization efficiency is strongly enhanced when dispersed within the LDH matrix. The LDH thus plays a critical promoting role, most likely by improving the coupling between CO₂ adsorption sites and NiCeO₂ hydrogenation sites, leading to enhanced ICCM performance of the hybrid materials. This interaction promotes both CO₂ adsorption and activation under ICCM conditions, thereby improving methanation efficiency. Although the present results clearly demonstrate a synergistic interaction between the NiCeO₂ and LDH phases, the available data do not allow the origin of this effect to be unambiguously identified. Factors such as improved NiCeO₂ dispersion, modified redox properties, increased surface area, and defect formation may all contribute to the observed synergy.

Previously reported LDH-derived Ni-based materials typically exhibit CH₄ productivities in the range of 0.24–0.33 mmol CH₄ g^{−1} cycle^{−1} [20,21,23,24], and under optimized conditions with relatively high Ni loadings (≈46 wt%), values up to 0.98 mmol CH₄ g^{−1} cycle^{−1} have been reported [22]. The NCL hybrids presented in this work have a comparable gravimetric CH₄ production (≈0.2 mmol CH₄ g^{−1} cycle^{−1}). However, when normalized to the active NiCeO₂ phase, the CH₄ productivity increases significantly, reaching up to 2 mmol CH₄ g^{−1} NiCeO₂ cycle^{−1}, corresponding to an order of magnitude enhancement in active-phase efficiency. These results demonstrate that the NCL hybrid materials presented here perform comparably and, in terms of active-phase efficiency, potentially surpass previously reported Ni-based DFMs. This improved performance reflects a more effective utilization of the catalytic component within the hybrid architecture, enabled by synergistic capture–conversion interactions rather than by increased metal loading alone.

3.3. Operando XANES during CO₂ capture–conversion cycles

To gain deeper insight into the active state of Ni during cyclic CO₂ adsorption–hydrogenation relevant to ICCM operation, operando XAS measurements were performed on the NCL₈ material. The Ni K-edge XANES spectrum (Fig. 6a) recorded at 100 °C after O₂ calcination at 450 °C exhibits a sharp white line and a pronounced 1 s → 3d pre-edge feature, characteristic of Ni²⁺ in a NiO environment [28,45,46,48]. The

presence of this oxidic phase is further confirmed by the FT-EXAFS spectrum (Fig. 6b), which displays contributions from Ni–O and Ni–Ni single-scattering paths typical of bulk NiO at phase-uncorrected distances of approximately 1.6 and 2.6 Å, respectively. Upon H₂ pretreatment, a substantial modification of both the XANES and FT-EXAFS profiles is observed (Fig. 6a and b), indicating the reduction of NiO to metallic Ni⁰. The FT-EXAFS spectrum collected at 300 °C is dominated by intense features at phase-uncorrected distances of ≈2.0 and 3.8 Å, assigned to the first and second Ni–Ni coordination shells, respectively, consistent with the formation of bulk metallic Ni particles.

After H₂ pretreatment, the catalyst was subjected to repeated He–CO₂–He–H₂ cycles to probe the evolution of the Ni phase under cyclic CO₂ adsorption–hydrogenation conditions representative of ICCM operation. The Ni K-edge spectra recorded at the beginning and at the end of the ICCM experiment (Fig. 7a) indicate that nickel remains in a mixed Ni⁰/Ni²⁺ state, while the corresponding difference spectrum (ΔXANES) suggests a net Ni oxidation with increasing the number of cycles. The XANES spectra collected continuously during CO₂/H₂ cycling (Figs. 7b and S8) display clear periodic variations in the pre-edge and white-line regions, reflecting reversible changes in the Ni state. In parallel, the mass spectrometry (MS) data shown in Fig. 7d reveal a sharp exchange between CO₂ and H₂ in the reactor, and the appearance of a CH₄ signal upon H₂ introduction confirms the hydrogenation of previously adsorbed CO₂. The detection of CH₄ under these conditions validates the operando nature of the XANES measurements, demonstrating that the spectroscopic features directly reflect catalyst behaviour under working ICCM conditions.

To quantitatively follow the evolution of nickel species during cycling, Linear Combination Fitting (LCF) of the Ni K-edge XANES spectra was performed using metallic Ni and NiO reference spectra. This approach enables deconvolution of the experimental spectra into relative contributions of Ni⁰ and Ni²⁺, providing time-resolved concentration profiles of each species. The resulting profiles (Fig. 7c) reveal pronounced oscillations in the NiO/Ni ratio that closely follow the imposed gas sequence (Fig. 7d). A direct comparison of the time-aligned LCF-derived concentrations with the MS signals (Fig. 7c,d) shows that the catalyst undergoes partial oxidation to NiO during CO₂ exposure, followed by reduction back to metallic Ni upon switching to H₂. This reversible redox cycling demonstrates that Ni in NCL₈ remains highly dynamic under ICCM conditions, with the present operando results providing direct, time-resolved evidence of this continuously evolving Ni⁰/Ni²⁺ redox behaviour.

These observations point to a dynamically evolving Ni phase under reaction conditions, with oxidation during CO₂ exposure and reduction under H₂. Such reversible Ni⁰/Ni²⁺ transitions are consistent with the behaviour expected for dual-functional materials under alternating

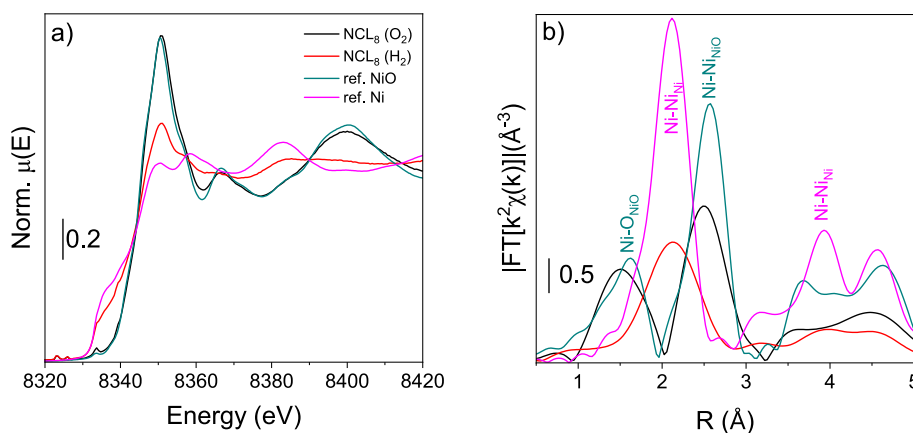


Fig. 6. (a) Ni K-edge XANES and (b) FT-EXAFS spectra collected on NCL₈ at 100 °C after calcination (black line) and after reduction at 300 °C (red line). Reference NiO and Ni metal spectra are included for comparison.

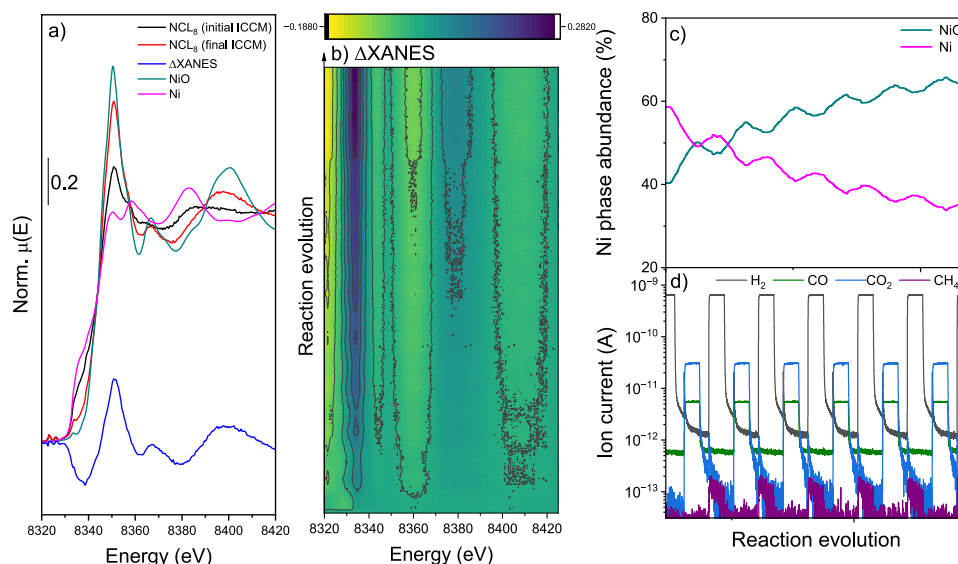


Fig. 7. Operando isothermal Ni K-edge (a) XANES and (b) Δ XANES spectra collected during He–CO₂–He–H₂ cycles. (c) NiO/Ni concentration evolution during cycles estimated by LCF. (d) Mass spectrometry data collected during cycles. Full operando XANES dataset is reported in Fig. S8. Reaction conditions for XAS experiment: 300 °C, 24 mg catalyst, CO₂ adsorption (5% CO₂/He, 40 mL min⁻¹, 5 min), He purge (40 mL min⁻¹, 4 min), H₂ hydrogenation (20% H₂, 40 mL min⁻¹, 5 min), and final He purge (40 mL min⁻¹, 4 min).

feeds, but have mainly been reported for high-temperature ICCU processes (~600 °C), particularly under RWGS conditions leading to CO formation [4,49]. In contrast, under typical methanation conditions (300–400 °C), Ni is generally considered to remain predominantly metallic [50,51], although partial oxidation of Ni has been reported during ICCU experiments at 350 °C in the presence of H₂O- and O₂-containing effluent gases [52]. Oxidized Ni species are considered inactive and commonly linked to catalyst deactivation rather than to the reaction mechanism [8]. In this context, the observation of a significant NiO fraction together with a progressive net oxidation of Ni during cycling, without any measurable loss in CH₄ production shown in Fig. 7c and d, may appear counterintuitive. This can be rationalized by the bulk sensitivity of XANES, which reflects the average oxidation state of Ni across the material. The formation of nickel aluminates as the origin of the observed Ni oxidation is highly unlikely, as their formation generally requires prolonged treatment under oxidizing atmosphere and temperatures above 500 °C [53], conditions very different than those employed here. Rather, the progressive increase in the average Ni oxidation state may be related to the high oxygen mobility of ceria [54]. During the CO₂ step, oxygen supplied by ceria may progressively oxidize bulk Ni species strongly interacting with the ceria matrix. Their complete re-reduction may not be achieved during the relatively short H₂ half-cycles at 300 °C, particularly since the complete reduction of bulk NiO species on ceria requires temperatures above 400 °C [30], leading to the gradual accumulation of oxidized Ni species within the particle interior. The detected oxidation is therefore likely dominated by changes occurring within the bulk of the particles, whereas the surface Ni sites, and/or the Ni–ceria interfacial sites, responsible for H₂ activation and methanation remain largely metallic due to rapid reduction under H₂. This surface–bulk redox differentiation, with a metallic active surface and a partially oxidized bulk, explains the spectroscopic observations alongside the preserved catalytic performance. Furthermore, an approximate estimation based on the operando XANES results indicates that the H₂ consumed for NiO reduction represents only approximately 1.5% of the total H₂ supplied during the H₂ half-cycle (see Supporting Information for details). These results suggest that H₂ consumption associated with NiO reduction remains a minor contribution during ICCM cycling.

3.4. Interplay between catalytic efficiency and environmental impact

The ICCM results reveal a clear interplay between CH₄ productivity and NiCO₂ catalyst utilization. Although NiCeO₂ achieves up to 30% higher CH₄ production when normalized to total material mass, the NCL hybrids produce up to seven times more CH₄ when normalized to the NiCeO₂ mass, indicating a more efficient use of the active phase upon integration with the LDH matrix. Consequently, larger quantities of NCL are required to achieve CH₄ outputs comparable to those of NiCeO₂. This trade-off between productivity and material demand motivates a comparative Life Cycle Assessment (LCA) to determine which configuration is better suited for large-scale application. The LCA quantifies and compares the environmental impacts of both systems (i.e. NiCeO₂ and NCL) and identifies the key contributors to the overall sustainability, providing insight into whether the efficient utilization of NiCeO₂ in NCL hybrids offsets their higher material requirements and production complexity.

3.4.1. Life cycle impact assessment of NiCeO₂ and LDH synthesis

The environmental impact of producing 1 kg of NiCeO₂ across 18 impact categories is shown in Fig. 8a, while the process contribution network and the corresponding Life Cycle Impact Assessment (LCIA) values are reported in Fig. S9a and Table S4, respectively. A key outcome of this analysis is that the dominant environmental burden does not originate from the Ni and Ce mineral precursors themselves, but rather from the energy demand (i.e., electricity mix), followed by the solvents used during synthesis (methanol and oleylamine). The contribution of electricity, here based on the European electricity mix employed in the baseline LCA, is particularly pronounced in water-related categories, including water consumption (WC) and freshwater eutrophication (FE). In contrast, the solvent use contributes more strongly to impact categories with greater variability, such as land use (LU) and marine eutrophication (ME). These trends mainly arise from the high electricity requirements of the synthesis process, as well as from ammonia consumption during oleylamine production and energy-intensive filtration and washing steps.

If the contribution of the electricity mix dominates the production of NiCeO₂, in the case of LDH it becomes even more decisive, representing by far the largest share among all inputs across nearly all impact categories. As shown in Fig. 8b, electricity accounts for more than 90% of the

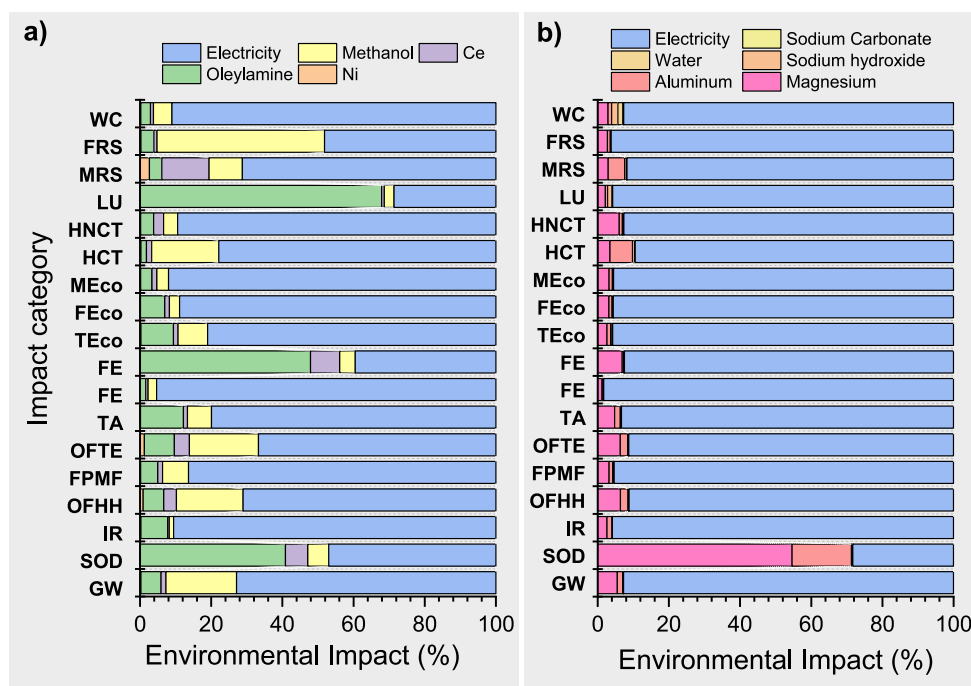


Fig. 8. Relative contributions of individual inputs associated with the production of (a) 1 kg of NiCeO₂ and (b) 1 kg of LDH across 18 ReCiPe 2016 Midpoint (H) impact categories. Key (top to bottom): WC, water consumption; FRS, fossil resource scarcity; MRS, mineral resource scarcity; LU, land use; HNCT, human non-carcinogenic toxicity; HCT, human carcinogenic toxicity; MEco, marine ecotoxicity; FEco, freshwater ecotoxicity; TEco, terrestrial ecotoxicity; ME, marine eutrophication; FE, freshwater eutrophication; TA, terrestrial acidification; OFTE, ozone formation (terrestrial ecosystems); FPMF, fine particulate matter formation; OFHH, ozone formation (human health); IR, ionizing radiation; SOD, stratospheric ozone depletion; GW, global warming.

relative impact in all categories, with the sole exception of stratospheric ozone depletion (SOD). This does not imply that electricity has a higher absolute impact for LDH than for NiCeO₂. On the opposite, the bar graph reports relative contributions, indicating that the influence of the other inputs is minimal for LDH. This trend is further illustrated by the process contribution network in Fig. S9b and by the comparison of the absolute LCIA values reported in Tables S2 and S3. For example, in the global warming (GW) category, the production of 1 kg of LDH results in 237 kg CO₂-eq (Table S4), whereas NiCeO₂ (Table S5) exhibits an impact of 14,459 kg CO₂-eq which is nearly 60 times higher. Aside from electricity, the next most significant contributions in LDH production arise from the magnesium and aluminum precursors, with magnesium nitrate (Mg(NO₃)₂) being the dominant contributor due to its approximately threefold higher loading relative to aluminum nitrate (Al(NO₃)₃) in the selected LDH stoichiometry.

3.4.2. Life cycle impact assessment of NCL preparation

As described in the experimental section, LDH and NiCeO₂ were mixed by ball milling using three different mass ratios to form the NCL hybrids. Accordingly, the energy demand of this mixing step was incorporated into the LCA, together with the impacts associated with the production of each constituent, weighted by their respective mass fractions in NCL. Based on the ICCM performance shown in Fig. 5, NCL₈ and NCL₁₆ were selected as the most relevant hybrids and benchmarked against NiCeO₂, which serves as the reference due to its highest CH₄ productivity per gram among all the studied materials. The functional unit was defined as the production of 1 mmol of CH₄. Because the NCL hybrids exhibit lower CH₄ productivity per unit mass than NiCeO₂, larger quantities of NCL₈ and NCL₁₆ are required to reach the same methane output. Accordingly, catalyst masses were adjusted in the LCA so that NiCeO₂, NCL₈, and NCL₁₆ correspond to identical CH₄ production, consistent with their ICCM efficiencies (Fig. 5). On this basis, 3.49 g of NiCeO₂ were compared with 4.97 g of NCL₈ and 7.73 g of NCL₁₆ as detailed in Table S6.

Fig. 9 compares the environmental impacts of NiCeO₂, NCL₈, and NCL₁₆ across the 18 LCIA categories, normalized by setting the highest value in each category to 100% and expressing the remaining materials relative to this maximum. The corresponding LCIA values are provided in Table S7. This representation reveals a markedly different ranking of materials compared with the CH₄ productivity trends in Fig. 5. While NiCeO₂ exhibits the highest activity per gram in the ICCM process, it also shows substantially higher environmental burdens in nearly all categories, calling into question its suitability for large-scale deployment. For an equivalent CH₄ output, NiCeO₂ displays impacts up to twice as high as those of the NCL hybrids in certain categories, notably marine eutrophication and land use. Overall, both NCL₈ and NCL₁₆ outperform NiCeO₂ from an environmental perspective, with NCL₈ showing a slight advantage over NCL₁₆.

To evaluate the reliability of these results, uncertainty was evaluated using a correlated Monte Carlo analysis (1000 iterations) comparing NCL₈ and NiCeO₂. As shown in Fig. S10, NCL₈ exhibited lower environmental impacts than NiCeO₂ in 100% of the simulation runs across all impact categories, demonstrating that this outcome is insensitive to uncertainty in the life-cycle inventory and impact assessment parameters. Thus, the Monte Carlo analysis confirms that the superior environmental performance of NCL₈ originates from intrinsic differences in materials, energy demand, and synthesis processes, rather than from stochastic variability, thereby strengthening the robustness of the comparative conclusions drawn in this study.

The LCA results above show that the electricity mix, here the European mix used in the baseline, dominates impacts in nearly all LCIA categories. A sensitivity analysis was therefore performed to assess the influence of different electricity mixes on the environmental performance of the most environmentally benign material, NCL₈, by comparing European (EU), French (FR), and global (World) electricity mixes. As shown in Fig. S11, the French mix yields the lowest impacts across most categories, reflecting its low carbon intensity and high share of nuclear and renewable energy. In contrast, the global mix leads to

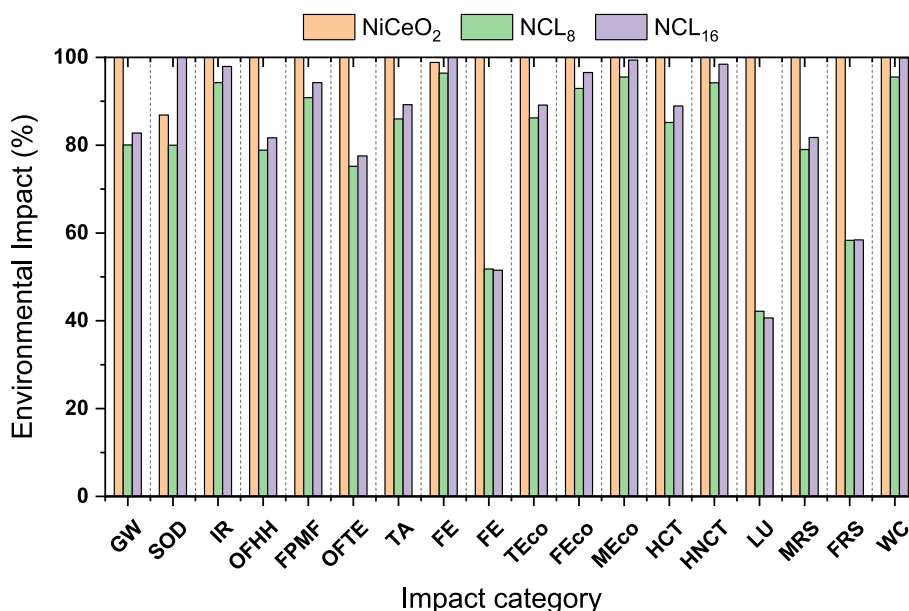


Fig. 9. Comparison histogram of the environmental impacts of NiCeO₂, NCL₈, and NCL₁₆ across 18 ReCiPe 2016 Midpoint (H) categories, normalized by setting the highest value in each category to 100% and expressing the remaining materials as percentages of this maximum. Results are reported for an equivalent functional unit of 1 mmol CH₄ produced.

substantially higher impacts due to its greater reliance on fossil fuels. Exceptions are observed for ionizing radiation and mineral resource scarcity, where the French mix performs less favourably because of nuclear power, and for water consumption and eutrophication, where the European mix shows higher impacts. Overall, these results highlight the strong potential to further reduce the environmental footprint of NCL production through cleaner, low-carbon electricity systems, emphasizing the benefits of transitioning toward renewable-based energy mixes.

4. Conclusions

Mixing NiCeO₂ catalyst with LDH ionic clay matrix provides an effective strategy to integrate CO₂ capture and methanation within a single material platform. The hybrids exhibit enhanced CO₂ uptake, stable low-temperature operation (300 °C) with almost 100% CH₄ selectivity, and excellent cyclic stability under ICCM conditions. While NiCeO₂ alone shows higher methane productivity per total catalyst mass, its integration within the LDH framework markedly improves active phase utilization, enabling up to seven times higher CH₄ production per mass of NiCeO₂. Operando XANES reveals reversible Ni⁰/Ni²⁺ redox transitions under alternating CO₂ and H₂ atmospheres, highlighting the dynamic nature of the active phase during cyclic adsorption–hydrogenation. Importantly, life cycle assessment demonstrates that this improved NiCeO₂ catalyst utilization translates into consistently lower environmental impacts at equivalent methane output. Together, these findings identify NiCeO₂/LDH hybrids as a scalable, noble-metal-free platform that bridges catalytic performance and sustainability for integrated CO₂ capture and conversion.

CRediT authorship contribution statement

Franck Morfin: Validation, Resources, Methodology, Investigation, Formal analysis, Data curation. **Javier Ivañez:** Investigation. **Ruben Checa:** Resources. **Laurent Piccolo:** Validation, Resources, Project administration, Methodology. **Anastasia Pappa:** Writing – original draft, Methodology, Formal analysis, Data curation. **Fengchen Zhou:** Investigation. **Thierry Dintzer:** Investigation. **Mathias Barreau:** Writing – original draft, Validation, Investigation. **Haroon Sumbhaniya:**

Investigation. **Samar Garreau-Hajjar:** Investigation. **Liva Dzene:** Writing – original draft, Methodology, Funding acquisition. **Davide Salusso:** Writing – original draft, Validation, Resources, Formal analysis, Data curation. **Spiros Papaefthimiou:** Data curation. **Spyridon Zafeiratos:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to check and improve the language of the manuscript. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2026.179101>.

Data availability

Data will be made available on request.

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